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GRE Data Interpretation

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1 Intro to Data Interpretation

The GRE will present a chart or graph with data, and then usually 3, sometimes 4, questions in a row will ask about that data.

GRE Data Interpretation tends to focus on real-world math, on math that arises in real-world scenarios: percents, statistics, ratios, etc.

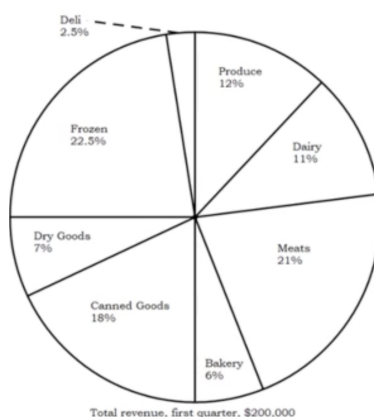
Some Data Interpretation questions demand a little more time than other quant questions, and you may have to allot as much as 2 minutes for the most challenging ones.

1.1 Strategies

- Understand the story of the graph.
- Check the units in the question: they may or may not match the data.
- Check the form of the answers before you perform your calculations.

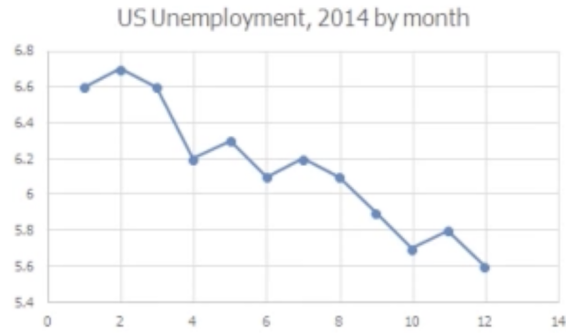
1.2 Types of Graphics

- Pie Charts



Typically, the GRE does not present questions that depend on only a single pie chart. Instead, pie charts are paired with another chart, and one breaks down a particular region of the other.

- Line charts



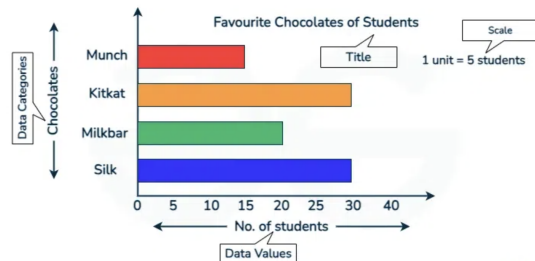
Almost always, the horizontal axis in a line graph is time. This is called a "time plot", and it demonstrates the trend over time.

- Bar graphs

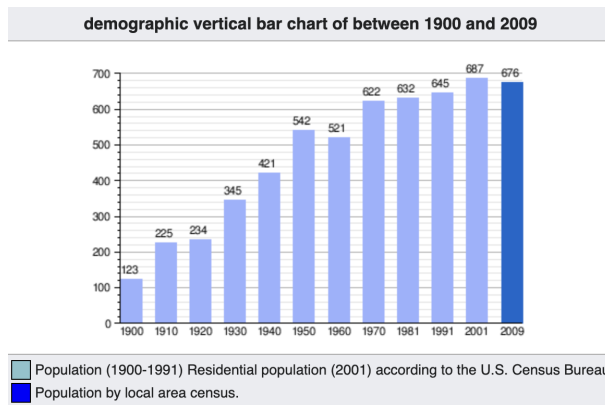
The bars may be vertical or horizontal.

The scale parallel to the bar indicates the variable in question, and the length of the bar indicates the amount or contribution of that element.

- Horizontal bars

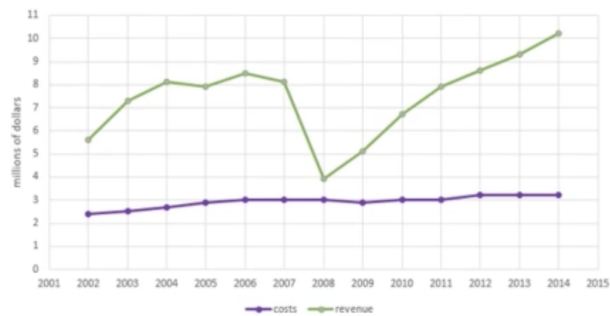


- Vertical Bars



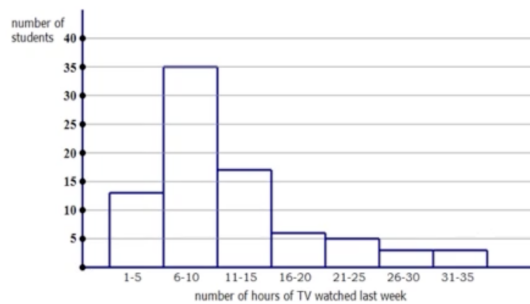
1.2.1 Examples

- The yearly costs and revenue for FabCo Corporation are shown.



Profit equals revenue minus costs. After 2004, in which year did FabCo's profit increase by the highest percentage from the previous year?

1. 2006
 2. 2008
 3. 2009
 4. 2010
- In a survey, 86 students were asked about their TV watching habits. Which of the following

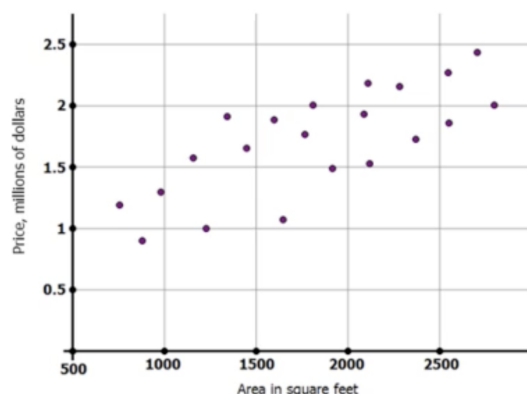


could be the median of this data set?

1. 5
2. 7
3. 11
4. 16
5. 21

1.3 Scatter Plots

Suppose, for a set of houses in a particular neighborhood, we measured both the price of the house and the area (sq. ft.) of the house: two numerical measurements for each house.



As we might expect, as area goes up, price tends to go up.

This pattern is known as a positive correlation: When one variable increases, the other one tends to increase.

Notice that correlation is about an overall pattern in the data set as a whole.

A correlation in the whole set does not mean that the pattern will be obeyed between every possible pair of points in the set.

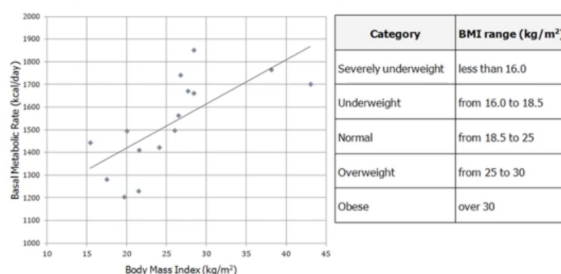
1. How many houses cost more than \$ 1 M and less than \$ 1.5 M?
2. How many houses have an area less than 1000 sq. ft.?
3. How many houses have an area more than 2500 sq. ft but cost less than \$ 2M?

A negative correlation is a pattern in which, as one variable increases, on average, the other decreases.

Sometimes, to demonstrate the trend of a correlation, a graph will include a regression line, also known as a "trend line" or a "best fit line".

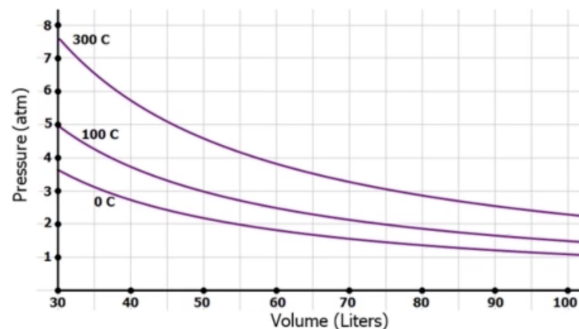
1.3.1 Question

- The graph below shows the BMI and BMR of 15 males between the ages of 43 and 65.



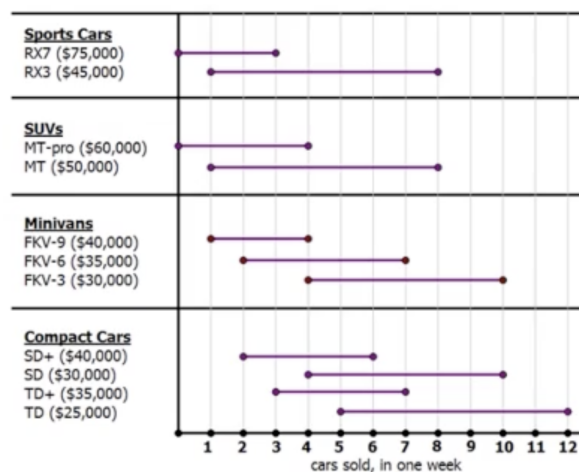
The trend line on the graph indicates, for a given BMI, what the expected BMR would be for the individual. In this group, individual have a BMR higher than predicted.

- The graph below demonstrates the relationship between volume and pressure for a sample of gas at three different temperatures: 0°C , 100°C and 300°C



Let P be the percent increase in volume at 3 atm, from 100°C to 300°C . Let Q be the percent increase in pressure at 75 liters, from 100°C to 300°C . P+Q is approximately:

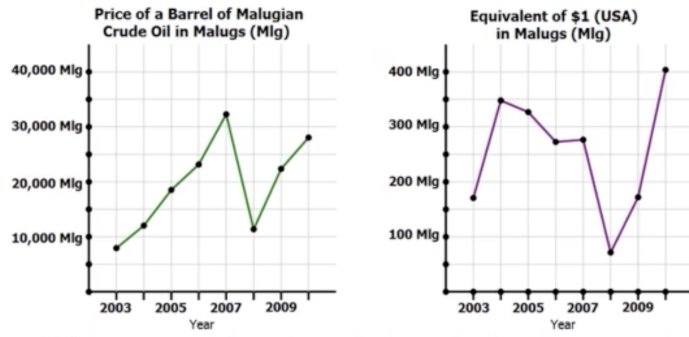
1. 25
 2. 50
 3. 75
 4. 100
 5. 125
- Over a two-year period, the Motorlux Car Dealership has kept track of the weekly sales of many of their models. The graph shows the range of these weekly sales for several models.



The minimum revenue from all compact cars combined is closest to how many times the minimum revenue from all sports cars combined.

1. 4
2. 6

3. 8
 4. 10
 5. 12
- The currency of the country of Malugia is malugs (Mlg). Malugia sells its own crude oil in this currency. The worth of \$1 (US) in malugs is shown on the graph on the right.



In which year was the price of a barrel of Malugian crude oil least expensive in US dollars?

1. 2003
2. 2004
3. 2007
4. 2008
5. 2010